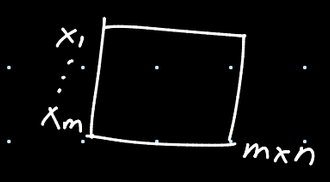
॥श्री गणेशाय नमः॥

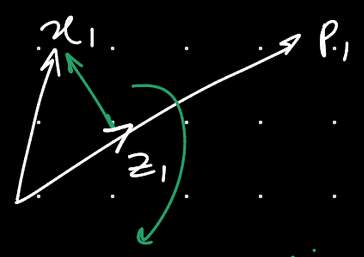
PCA :

- Approaches :
  - → maximize the projected variance.
  - → Get more uncorrelated features.
  - → minimize the reconstruction error.

Design matrix



- maximize variance :



$$P_1 = \underset{P}{\operatorname{argmax}} \frac{1}{m-1} \sum_{i=1}^{m} (P^T X_i - P^T \bar{X})^2 \quad \text{s.t. } \|V\|_2^2 = 1$$

P_1 ← 1st principal component

$$= \underset{P}{\operatorname{argmax}}\ P^T C P$$

C = covariance matrix.

Soln: $CP_1 = \alpha P_1$

$P_1 \Rightarrow$ eigenvector corresp. to α

$\alpha \Rightarrow$ largest eigenvalue of covariance matrix

$$\text{Proportion of variance captured} = \frac{\alpha_1 + \alpha_2 + \dots + \alpha_K}{\alpha_1 + \alpha_2 + \dots + \alpha_n}$$

- Uncorrelated features :

Principal components are linear combinations of the original features.

$$x_i = \alpha_{i1} P_1 + \alpha_{i2} P_2 + \dots + \alpha_{in} P_n$$

$$\alpha_{ij} = x_i^T P_j$$

$$\hat{x_i} = [\alpha_{i1} \ \alpha_{i2} \ \dots \ \alpha_{in}]_{1\times n}$$

$$\hat{X} = \begin{bmatrix} - x_1^T - \\ \vdots \\ - x_m^T - \end{bmatrix}_{m\times n} \begin{bmatrix} | & & | \\ P_1 & \cdots & P_n \\ | & & | \end{bmatrix}_{n\times n} = \begin{bmatrix} - \hat{x}_1^T - \\ \vdots \\ - \hat{x}_m^T - \end{bmatrix}_{m\times n}$$

← features in the transformed space

$$\hat{X} = XP$$

Covariance matrix

$$C(\hat{X}) = (\hat{X})^T (\hat{X})$$

$$C(\hat{X}) = P^T P$$

soln comes out the same $\Rightarrow$ P is eig val matrix

(squared projection distance)

- Reconstruction error :

$$P = \underset{P}{\operatorname{argmin}} \frac{1}{m} \sum_{i=1}^{m} \|x_i - \hat{x}_i\|_2^2 \quad \text{s.t. } P^T P = I$$

If data is centered, i.e. $\bar{x} = 0 \Rightarrow \hat{x}_i = P(P^T x_i)$

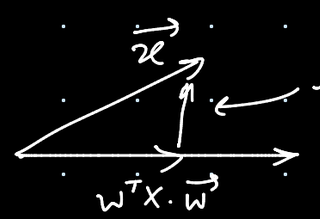
- SVD interpretation:

$X = U\Sigma V^T \Rightarrow X^T X = V\Sigma^2 V^T$ ← eigen vecs. of cov. matrix.

so direction of principal components is along the right singular vectors.

* Principal components are chosen so as to maximize variance.
* PCA reduces overfitting by removing poorly predictive dimensions.
* obviously computation becomes faster.
* PCA does NOT tend to reduce bias, but rather increases it. It decreases variance in error.
* These do NOT change the result of PCA:
  ① Appending a col^m of 1's to the design matrix.
  ② PCA from d-dim to j-dim then to k-dim. VS directly from d to k principal components.
  ③ Rota^n of all points as a set in the feature space changes. PCA, but does not change the largest eigenvalue of cofactor matrix.

* Reconstruction error: here w has principal components.



$$= \| \vec{x} - (w^T x)\vec{w} \|^2 = \| \vec{x} - \vec{w}(w^T x) \|^2 = \| X - WW^T X \|_2^2$$

* fraction of variance captured $= \dfrac{\lambda_1 + \lambda_2 + \dots + \lambda_K}{\lambda_1 + \lambda_2 + \dots + \lambda_n}$ variance of projected data $= \frac{1}{N} V^T X X^T V = \frac{\lambda}{N} V^T V = \frac{\lambda}{N}$ ($XX^T V = \lambda V$)

* covariance matrix:
  → All diagonal elements are +ve (since they represent variance).
  → Symmetric.
  →